Jason Mathew

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EDUCATION

University of Pennsylvania, School of Engineering and Applied Sciences, Philadelphia, PA — *May 2027*

Robotics, M.S. | Mechanical Engineering, B.S. — Minors: *Computer Science, Math* — *GPA*: 3.78/4.0

RELEVANT EXPERIENCE

Zoox | *Technical Program Management, Hardware Development* — *May 2026 – Present* | San Francisco, CA

- Launching first end-to-end development process for 3+ future vehicles, create dependency map across 40+ teams
- Delivered real-time visibility into engineering resource allocation for 400+ engineers by developing and deploying full-stack Python web application in <2 weeks. Automated workflow from data collection to reporting in interactive live dashboards.
- Lead testing qualification campaign for seatbelt pyrotechnic-tensioning enclosure, characterizing wiring resistance to meet design validation criteria for production release. Established electrical performance requirements for next-gen robots.

SpaceX | *Hardware Product Design, Falcon Avionics Core Engineer* — *May 2025 – August 2025* | Los Angeles, CA

- Redesigned common engine-controller enclosure (9 per vehicle) to be compatible with multiple PCB revisions; use Ansys to validate structural durability under 40x flights; increased build-production output 8% by removing unreliable features
- Predicted reliability of high-risk PCBAs using Ansys Sherlock to assess PCB component response to fatigue, shock, and vibe
- Lead critical booster component qualification to pass vibration and thermal fatigue; achieved buyoff in <6 weeks
- Investigated hardware reliability issue of tank RTDs/cameras; root cause analysis, CT scans, and Python data analysis
- Designed $30k cheaper, in-house dyno braking system to test high-voltage motors; accelerated project timeline by 1 month

Penn Electric Racing (FSAE) | *Mechanical Lead* — *May 2025 – May 2026* | Philadelphia, PA

- Technical Program Manager of 60+ engineers in 6 subsystems: drivetrain, battery, chassis, aerodynamics, suspension, vehicle dynamics. Set architecture to balance performance and reliability. Best season in 8 years; 15th overall, 6th in design.
- Design critical components across vehicle; cell module enclosure, steering system, drivetrain mounting, chassis tabs
- Manage electromechanical integration for powertrain modules, tractive system, wiring harnesses, and 30+ sensors
- Finished entire build 2 months faster than previous years with 80% more on-track testing and 150+ miles driven
- 40% faster fabrication time for complex assemblies from coordinating GD&T, integration, and DFM/DFA reviews in CAD
- Driver Systems Lead — *May 2024 – May 2025*
    - Placed top 10 in Design Event at competition with first principles-based design. See Design Brief Slide Deck
    - Owned design of pedals, brakes, steering, dashboard, ergo and managed cross-subsystem integration: performance tradeoffs, structural and thermal FEA simulations, CNC and plastic manufacturing for 40+ parts, testing validation

Precision Machining Lab Assistant | *Teaching Assistant* — *August 2025 – Present* | Philadelphia, PA

JHU APL, Force Projection Sector | *Autonomous Systems Design Engineer* — *September 2022 – August 2023* | Laurel, MD

- Secret Level Security Clearance, working on 3+ software projects for Guidance, Navigation, and Controls team
- Developed multiparameter design optimization model for aerodynamic systems in MATLAB & C++, reduced runtime 60%
- Contributed to MATLAB aerodynamic model for hypersonic airplanes, meshing and using panel methods to estimate Cp

PROJECTS

HMS Switch Interface | *Circuit Architect, Penn ADAPT* — *January 2024 – May 2024*

- Designed, routed, and soldered PCBs for switch interfaces—assistive technology for individuals with cerebral palsy
- Specialized in creating ultra-compact circuit cards to fit within a 1-inch x 2-inch x 0.5-inch enclosure
- Optimized enclosure box for ingress protection, impact resistance, non-flammability, and insulation properties

Programmable Intelligent Collaborative Engagement Munitions (PICEM) | *JHU APL* — *May 2023 – August 2023*

- Collaborated on a MATLAB optimization project, developing decision-making methods for non-centralized projectile swarms
- Worked closely with team to refine and integrate object-association and object-assignment algorithms
- Created testing framework to evaluate reliability of PICEM algorithms operating autonomously, tested 50+ edge cases

TECHNICAL SKILLS

- **Platforms/Tools:** Unigraphics/Siemens NX, Ansys, SolidWorks, Ansys Sherlock, MasterCAM, Jira, Confluence, Teamcenter
- **Skills:** Structural Simulation, PCB Simulation, CNC Machining, DFM/DFA/DFS, GD&T, Plastic/Composite Manufacturing